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Skin-integrated Haptic Interfaces enabled by Scalable Mechanical Actuators for Virtual Reality

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Abstract—The very recent concept of metaverse highlights the importance of virtual reality (VR) and augmented reality (AR), that associates with a wide variety of applications in entertainment, medical treatment, and human machine interfaces. The current VR/AR technologies mainly rely on visual interaction, while immersive experience in VR and AR highly demands sensational feedback, such as haptic and temperature with noticeable quality in wearable or even skin-integrated formarts. In this paper, we report a wearable and flexible haptic interface based on electromagnetic vibrotactile actuators with high wearability and stability. By adopting double layers of copper (Cu) coils at the top and bottom of the magnetic disc, an enhanced electromagnetic field can be generated. Additionally, the intensity of the haptic feedback can be modulated according to sensed pressure in the virtual world by adjusting the value of power input and frequency. The actuator exhibits high stability and tolerance upon environmental, cyclic, and impact resistance tests. Finally, the actuators are developed into the soft VR interfaces for mounting on forearms, fingers and hands to verify their superiority over conventional haptic actuators in the aspects of performance and applications.

Index Terms—Haptic feedback, electronic skin, harsh electronics, VR, actuators

I. INTRODUCTION

Haptic interfaces in wearable formats reproduce tactile sensations on the skin, enabling applications in the fields of virtual reality (VR)/augmented reality (AR), education, entertainment, and medical prosthesis[1-10]. The recent concept of metaverse associating the interactions between users and virtual worlds highlights the importance of haptic feedback as which can provide a vivid sense of realism to users[1, 11, 12]. For instance, users wearing haptic devices

can get immersive experiences in VR applications by feeling tactile information of the graphical virtual materials[13-15]. There are various feedback technologies serving for haptic interfaces, such as actuators based on electromagnetic field[1, 6], electrostimulation[16-18], piezoelectric effect[19], and direct current (DC) driven offer-center motors[20]. Some of these devices have been commercialized and used in personal electronics, however, most of them are still facing hurdles of wearability for long-term applications[5].

Some efforts have been made to develop wearable haptic interfaces, specifically technologies of electrostimulation and vibration-based mechanical actuators[7, 21]. The major challenge of electrostimulation is the sensitive thresholds which are quite different among individuals, limiting the universal applications[22-25]. While a promising alternative option in generating haptic sensation is using mechanical actuators, as which provide physical force to users[1, 24, 26]. Despite the advantages of mechanical actuators, there are still a lot of room of the haptic interfaces can be improved by considering the wearability, miniaturized structure, and operational stability of the state-of-art mechanical actuators[1, 4, 5, 27]. For example, the actuators we previously reported adopted a cantilever beam structure, where the magnet was exposed outsides, that may result in the low operational stability in ambient or extreme conditions, such as humid environment[1]. Furthermore, the requirement of high density energy often results in using bulky batteries or wired power sources[27]. Therefore, a certain magnitude of response is required to notify users from interaction successfully. Some recent works reported actuators with flexible features, but also faced the hurdles of complicated fabrication processes and

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fragile structures, which could undoubtedly bring extensive uncertainty in practical applications[2]. In addition, less considerations of adopting scalable actuators with a high channel of counts arrays also result in challenges in the accuracy and fidelity of feedback over different parts of body [4, 28, 29]. As the key component for next generation haptic feedback interface, the mechanical actuators should exhibit great advantages over the existing technologies as follows: (a) capable of providing wide-range pressure with adjustable vibration frequency for users. (b) robust structure with excellent mechanical design for operating normally in various extreme environments, such as water, sand, and high-temperature weather. (c) scalable sizes and array densities so that they could generate accurate haptic feedback to users' different skin areas for an immersive experience.

Herein, we introduce miniaturized, wearable vibrotactile actuators that are capable of operation in harsh environments for unconventional applications in immersive VR and AR. Distinguished from the reported electromagnetic mechanical actuators, the actuators in this work adopt a bilateral design, that are named as bilateral vibrotactile actuators (BV-actuators), as the vibration intensity can be significantly enhanced through the coupled electromagnetic field induced by two Cu coils (**Fig. S1** and **Fig. 2a**). As a result, the BV-actuators exhibit advantages from many aspects, from scalability, to size, to output performance and to stability in various working environments, comparing to other available vibrotactile actuators (**Table. S1**). Therefore, the actuators offer benefits in designing a compact wearable VR interface due to its low-power consumption characteristic. With a working frequency range of 10~150 Hz, the actuators could well stimulate the mechanoreceptor (Pacinian corpuscle) with a maximum yielded pressure of 0.72 kPa [30]. In addition, the BV-actuators possess great stability and mechanical tolerance under various operating environments and working conditions. The BV-actuators are functional in air, water, and sand, and it is capable of withstanding a cyclic test with 400,000 iterations in air without failure. Furthermore, the results of the mechanical tolerance measurements verify that the BV-actuators could survive under strong crashing with a pressure of ~5.53 MPa. The simple device design allows the BV-actuators to be configurable from mm to cm in demission, and thus enables functions on various parts of the body based on the skin sensitivity variations over regions. To provide excellent feedback performance of haptic interfaces based on the BV-actuators array, high density actuators are demonstrated by the integration with flexible printed circuit board (FPCB) and silicone polydimethylsiloxane (PDMS) encapsulation. The wearable VR interfaces can successfully offer programmable haptic feedback to users in VR games, and thus enable users to experience immersive virtual reality with dynamic and vivid haptic feedback.

II. RESULTS

A. Overview of the BV-actuator

Fig. 1a shows the schematic illustration of the applications for the BV actuators embedded wearable VR interface, in which a user wears a VR headset and the haptic feedback system interact with the gaming graphics. **Fig. 1b** shows the schematic diagram of the soft haptic interface with a high channel of counts BV actuators array, where two thin soft PDMS layers (170 μm , 67 kPa) serve as the top and bottom substrates/encapsulations, an FPCB with surface mounted microelectronics and metallic traces serve as the circuit and interconnects. In the FPCB, conductive copper (Cu) circuits embedded in a highly bendable polyimide (PI) sheet, that allows to protect the circuit from potential damage and short circuit (**Fig. S2**), while the arrays of the BV-actuators are attached on the FPCB by soldering. Controlling the electrical input of the flexible electronics allows the actuators to form programmable force distributions on the skin. Two thin and soft PDMS layers encapsulated the device at the top and bottom can further protect the electronic components from damage and thus provide a conformable skin interface. As a result, the whole device is immensely bendable and twistable (**Fig. S3** and **S4**) and can be attached to the human epidermis tightly without delamination during the user's activities. **Fig. 1c** shows the structural design of the BV-actuator, where the device is composed of six layers. An adhesive tape at the bottom acts as the substrate of the actuator. A 3D printed epoxy ring at the middle with top and bottom of Cu coils/polyethylene terephthalate (PET) serve as the bone structure of the actuator. The two Cu coils can generate coupled magnetic fields when electrical inputs are applied, and thus the permanent magnet disc in the chamber moves up and down induced by Lorentz force, performing as a vibration element. The two layers of PET between the coil and the magnet serve as a protection layer to avoid the direct contact between the magnet and the two Cu Coils, and also act as sealing layers to enable waterproof properties. By modulating the current directions in two Cu coils, the actuators could yield a higher pressure with the opposite current directions in the two coils (**Fig. 2a**). The simple device layouts afford a configurable fabrication procedure for the actuators from mm to cm scale. Here we demonstrate 3 sizes of the device with diameters of 3.5, 6.5, and 11 mm (thicknesses of 2.8, 3.2, and 3.6 mm respectively) (**Fig. 1d** and **Fig. S5**). Thus, depending on the variations in the spatial distribution of the mechanoreceptors under the different parts of the body, actuators in arrays with appropriate scales can be chosen to maximize the sensation and resolution of the feedback provided by the haptic interface. **Fig. 1e** shows the photo of an actuator with a size of 11 mm $\times$ 2.8 mm (diameter $\times$ thickness) in the top and side view respectively.

To test the skin sensitivity to the BV-actuators, we measured the minimum distances between two neighbor actuators for discrete haptic feedback at three volunteers' different body regions (finger@ 5.33 mm, palm@ 6 mm, hand back@ 6.83 mm, forearm@ 11 mm, upper arm@ 25mm, shoulder@ 37.33 mm, chest@ 19 mm, back@ 75 mm, and calf@ 19 mm), that can help to realize guidelines in the actuator arrays design, including parameters of the actuator size and

gaps among them (**Fig. S6** and **Fig. 1f**). As shown in **Fig. 1f**, measurements of skin sensitivity of 9 body locations offer guidelines of designing the haptic interface and also demonstrate the great potentials of the actuators as the wearable VR interface. As a result, a relatively more compact design could be applied for fingers, and the back is the most insensitive part. Considering the location changes of the actuators due to the bending of the haptic interface during use, mechanical and electrical couplings are two critical issues in the device designing. To obtain the minimum distance between two adjacent actuators, we measured the pressure output of a working actuator by a flexible pressure sensor (**Fig. S10**), where the coupling was set by changing distance of the tested actuator with another in four directions (**Fig. S7**). The minimum distance between two actuators is determined by continuous monitoring the sensing signal of the pressure sensor by moving one actuator to the other one. Once obvious signal fluctuation or decrease of the pressure sensor was observed, the distance between the two actuators can be considered as the minimum one, which is capable of guiding the design to avoided coupling. **Fig. S7b** shows the measured minimum distances for the four situations. Based on the data listed in **Fig. 1f**, wearable BV-actuator based VR interfaces targeting fingers, palms, and forearms were designed and fabricated (**Fig. 1g**). Since fingers are the most sensitive part, two different haptic interfaces with two actuator sizes (diameters of 3.5 and 6.5 mm) and density (3×3 and 2×2) were selected. Actuators with a diameter of 11 mm were exploited for the haptic interfaces used for the palm and forearm. The soft nature of the materials and devices used in the haptic interfaces allow the electronics to be tightly mounted on skin (**Fig. 1g**).

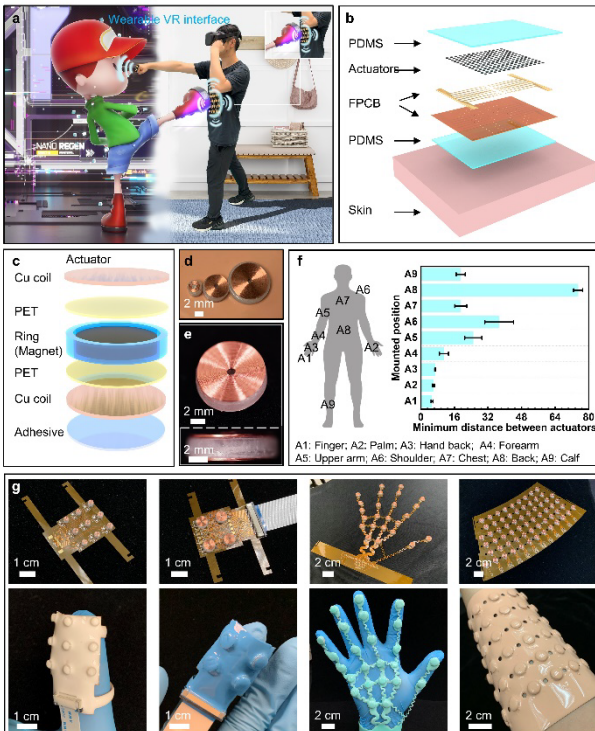


Figure 1. Overview of the BV-actuator and its applications. (a) Optical illustration of the wearable VR interfaces. (b) Exploded structure of the BV-actuator wearable VR interface (c) Structural illustration of the BV-actuator. (d) Optical image of the actuators with different diameters of 3.5, 6.5, and 11 mm, respectively. (e) Optical images of the actuator, top and side view. (f) Shortest distance for discrete haptic feedback between the actuators at different body parts. (g) Optical images of the BV-actuator wearable VR equipment targeting a finger, palm, and forearm.

B. Operational performance of the BV-actuator

To investigate the optimal structural dimensions and vibration frequency of the BV-actuators, the performance of the device had been systematically studied by conducting experiments with controlled variables. **Fig. 2a** shows the working mechanism of the BV-actuators under alternating current (AC) electrical power input. Following the right-hand rule, AC power applied to the Cu coils generates magnetic field to the actuator, thus the vibrating magnet disc is trapped between the two layers of coil/PET, and thus vibrates at same frequency of the AC inputs. The Cu coil with electrical power produces a static magnetic field around itself, which forms the shape of a bar magnet giving a distinct North and South pole. The magnet between the top and bottom coils is subject to forces in the same direction when the current flows in opposite directions in the two coils (**Fig. 2a i**); otherwise the magnet is subject to opposite forces when the current flows of coils in the same directions (**Fig. 2a ii**). With the presence of two separate magnetic fields generated by the two Cu coils, a much stronger vibrational force can be achieved compared to similar actuators with a single Cu coil (**Fig. 2a iii** and **S8**). **Fig. S9** shows the pressure generated by the actuators (shown in **Fig. 2a**) under various conditions with a constant AC input of 63.75 mA and 60 Hz, which further proves that the actuator could yield the most intense vibration under the condition i.

As the actuator generated high- and low-pressure force alternatively, clear vibrations can be delivered to a user through rhythmical pressure difference. In the experiments, the actuator was directly fixed on the self-developed flexible pressure sensor to record the numerical strength of pressure generated by the vibrations produced by the device during operation (**Fig. 2b**). As shown in **Fig. S10**, the BV-actuator on the sensor was operated by AC power input supported by a waveform generator while the pressure measured was recorded and displayed on a computer. The self-developed sensitive piezoelectric pressure sensors composed of two thin layers of platinum (Pt) with a size of $1 \text{ cm} \times 1 \text{ cm} \times 50 \text{ nm}$ (length $\times$ width $\times$ thickness) enveloped barium-calcium zirconate titanate (BCZT) with a size of $1 \text{ cm} \times 1 \text{ cm} \times 400 \text{ nm}$ (length $\times$ width $\times$ thickness) and two layers of polydimethylsiloxane (PDMS) with different thicknesses (50 μm at top and 1mm at bottom) encapsulated the composite of two Pt and one BCZT layers for protection produces different values of output voltage against pressure force applied on due to piezoelectric effect (**Fig. S11**). Furthermore, to optimize thickness of the actuators with a diameter of 11 mm to produce the strongest vibrational force,

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the devices with three thicknesses of the resin ring (1.9 mm, 2.3 mm, and 2.7 mm respectively) had been tested at different vibration frequencies (from 10 to 150 Hz with an increment of 10 Hz) under uniform power input. As shown in **Figs. 2c, 2d, and 2e**, the actuator with the ring thickness of 2.3 mm generates the most intense feedback (0.72 kPa at 60 Hz) while the device with the ring size of 2.7 mm actuator delivers (0.52 kPa at 50 Hz) a stronger force than that of 1.9 mm actuator (0.43 kPa at 70 Hz), where the R-square values from the fitted curves for each actuator is 0.94, 0.98, and 0.96 respectively. Following, the actuator showed the best performance with the ring thickness of 2.3 mm (overall thickness, 3.2 mm), which may result from that 2.3 mm is an optimal travelling distance for the magnet to achieve a highest kinetic energy for generating the highest pressure (0.72 kPa). And the optimum operating frequency of the device was verified to be 60 Hz (**Fig. 2d and S12**). The steady pressure output induced by the actuators demonstrate the controllable and accurate haptic feedback (**Fig. S12**). After the verification of the optimal thickness (3.2 mm) and operational frequency (60 Hz), we studied the electrical power input effect on the actuator, where the ranges of vibrational force produced based on different AC input were recorded by applying 12.19 mA (0.29 V), 16.75 mA (0.40 V), 21 mA (0.50 V), 24.63 mA (0.59 V), 28.75 mA (0.69 V), 33.75 mA (0.81 V), 40.75 mA (0.98 V), 49.38 mA (1.19 V), and 63.85 mA (1.53 V), respectively. As shown in **Fig. 2f and 2g**, as the quantity of input current elevates, higher levels of pressure had been measured obviously (0.17 kPa, 0.24 kPa, 0.30 kPa, 0.36 kPa, 0.45 kPa, 0.49 kPa, 0.59 kPa, 0.63 kPa, and 0.72 kPa, respectively). It is proved that the magnitude of haptic feedback can be regulated by adjusting the power input according to the sensed stimulus. Thereafter, the relationship between operating performance and diameter of the actuator can be demonstrated by applying increasing AC input to the actuators with diameters of 3.5, 6.5, and 11 mm at a constant thickness of 3.2 mm. **Fig. S13** shows the pressure outputs of the two actuators with diameters of 3.5 mm and 6.5 mm as a function of applied AC power frequency. The resonant frequency is ~50 Hz, which approach 60 Hz. Here, we adopt 60 Hz as their working frequency with two considerations: (1) the pressure outputs at 60 Hz are approaching to that at 50 Hz for the two actuators. (2) same working frequency for the three actuators could share a control panel for later manipulating actuators arrays, simplifying the circuit design. According to **Fig. 2g**, the actuator with the larger diameter can generate noticeably a wider range and higher value of haptic feedback while more sharp increases are observed simultaneously. Furthermore, a larger quantity of power input can be managed by bigger actuators. As a result, the larger diameter BV-actuators can be applied on target body parts with less sensational sensitivity. Furthermore, through the power input effect, it is verified that the magnitude of the vibrational force generated by the actuator can be manipulated by regulating the electrical power input, its dimensions and AC frequency. **Fig. S14** shows the electrical performance of a commercial actuator based on piezoelectricity (Beijing Ideal Technology Co., Ltd.) with a size of 6 cm × 3 cm,

and it is obvious that our actuators could yield a higher pressure to users, contributing to a stronger haptic feedback. **Fig. S15** shows the theoretical results of the 11-mm actuator working on an artificial skin (PDMS, 147 kPa) with a generated pressure of 0.17 kPa, resulting in a maximum displacement of the platform around 0.19 μm. The displacement value (0.19 μm) is much larger than the skin strain threshold induced by vibration (0.01 μm)[30], demonstrating that our actuator could make sensations on human skin.

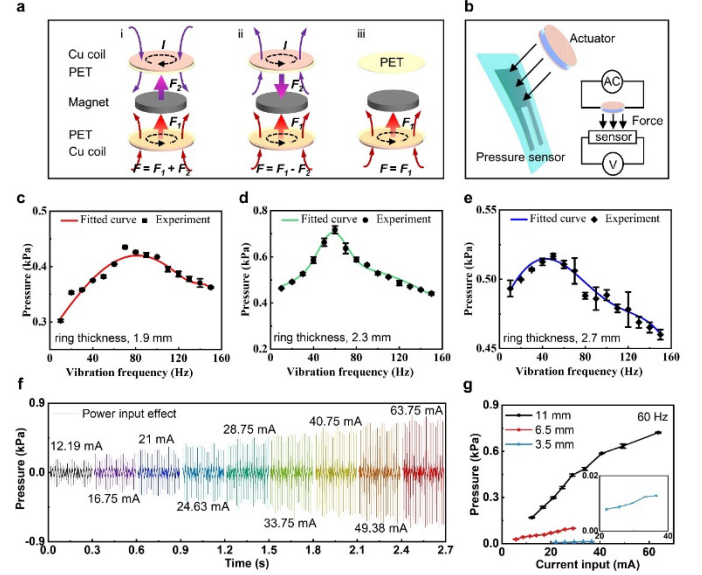


Figure 2. Experimental results illustrating working mechanism and operational performance of the BV-actuator. (a) Schematic illustration of working procedures of the BV-actuator. (b) Schematic illustration of the experimental setup for measuring the pressure induced by the device. Operational performance of the BV-actuators with different ring thicknesses of 1.9 (c), 2.3 (d), and 2.7 mm (e) respectively under increasing vibrational frequency. (f) Operational response of the device with increasing power input under 60 Hz of vibrational frequency, as a function of time. (g) Operational performance of the BV-actuators with different ring diameters of 3.5, 6.5, and 11 mm under increasing electrical power input.

III. TOLERANCE CAPACITY

Next, the impact resistance capacity and the thermal characteristic of the device were also evaluated to verify its stability and operation guarantee under various environmental conditions. Uniformly distributed force with pressure of 5.53 MPa was applied on the top surface of the actuator in FEA, where the PET reaches its compressive yield strength (40 MPa) as shown in **Fig. 3a**. The stress distribution of other parts of the actuator including Cu coil and epoxy were calculated in **Fig. S16**. Although the maximum stress in copper coils also achieves up to 93 MPa, much larger than the yield strength of copper, the position of the maximum stress is located at the position where the copper overlaps with epoxy, and the coils will not break down as the maximum stress is far away from the ultimate tensile strength of copper. As a result, the actuator could operate normally under the 5.53 MPa. Subsequently, an

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impact resistance test was performed for checking its tolerance against mechanical impact. **Fig. 3b** illustrates the experimental setting of the test that a tester wears a nitrile glove to continuously smash the actuator for over 100 times. Before and after the resistance experiment, the actuator is capable of generating a constant vibrational force without any decay, demonstrating its excellent stability (**Movie S1**). Subsequently, a cyclic test was conducted to further prove the repeatability of the actuator (**Fig. 3c**). The experiment iterated 400,000 times in total while the conditions and variables were maintained constant. Consequently, under fixed conditions (the overall thickness of 3.2 mm, vibration frequency of 60 Hz, and input current of 40.75 mA), the actuator successfully verified its stability and repeatability by inducing constant vibration intensity during the cyclic experiment. Thereafter, the actuator was operated in air, water, and directly on sand respectively as it was placed on a PDMS substrate (PDMS: crosslink = 30:1) with a size of 7.5 cm × 7.5 cm × 3 mm (length × width × thickness) for mimicking human epidermis to validate the workability of the actuator under different environments (**Fig. S17, Movie S2, S3, and S4**). We used a thermal imaging camera to capture and quantify the maximum temperature, as shown in **Fig. 3d, f, h** (61.6 °C, 29.6 °C, and 69.2 °C respectively), which is further proved by the thermal theoretical results (**Fig. 3e, g, i**). For this experiment, the actuator was set to the maximum vibration force (63.75 mA at 60 Hz). Through **Fig. 3j**, the maximum thermal dissipation of the BV-actuator under different ambient environments are compared, where the peak temperature (69.2°C) generated by the actuator is the largest on the sand and it was slightly lower than in air, while it shows the lowest temperature in water. In addition, the maximum temperature (69.2°C) is much lower than the magnet Curie temperature (320~380°C), demonstrating that the actuator could stably operate without concerns on the magnet performance. As the threshold temperature for an instant skin burn is approximately 60 °C, the current input and time required for the actuator to reach the maximum temperature in air was measured under fixed variables (60 Hz of vibration frequency and 63.75 mA of input current respectively). Referring to **Fig. S18**, under 60 Hz, the actuator produced the peak temperature when 63.75 mA of electrical power was applied. Following, under 63.75 mA, more than 1 min was taken for generating the peak temperature (**Fig. 3k**). Furthermore, approximately 25 seconds were taken for reaching 45°C which is the thermal point where human commences to feel a burning sensation. As a result, the minimum duration for reaching the maximum thermal energy level was revealed as more than 1 min that is a fairly long-time in the aspect of haptic feedback initiated by a touch. **Fig. S19** shows the thermal dissipation process of the actuator with the initial temperature of 69°C, and it requires over 6 mins to cool down the actuator. Finally, the actuator was operated under high and low temperatures in air (60 and 1.5 °C), in water, and in sand respectively to test its thermal tolerance and to firmly confirm the working environments of the BV-actuator (**Fig. 3l**). As a result, it was proved that the actuator is able to generate vibrations without any degradation under

ambient conditions. Consequently, via experiments, working environmental variations (in air, water, and on sand) and its high stability and tolerance against mechanical impact were verified. In addition, the potential risk of skin burn resulting from the actuator could be investigated that strongly recommends users not to be exposed to the maximum vibration force of more than 1 min.

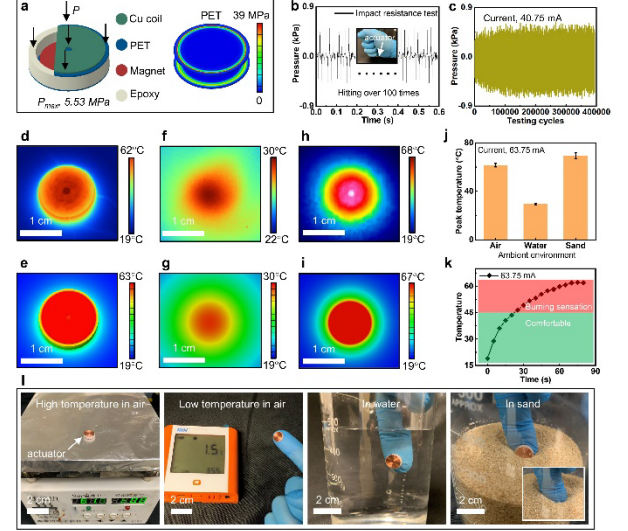


Figure 3. Operating characterization and tolerance capacity of the BV-actuator in the aspects of mechanical impact and heat. (a) FEA model of the actuator under uniformly distributed force with pressure of 5.53MPa and the stress distribution of PET component. (b) Operational performance of the BV-actuator before and after over 100 cycles of impact resistance test with optical image of the experimental setup for impact resistance testing. (c) Cycling performance of the BV-actuator under power input of 40.75 mA for 400,000 testing cycles. Thermal and FEA images of the device in the air (d, e), water (f, g), and on sand (h, i) respectively during operation. (j) Maximum temperature generated by the BV-actuator under different environments. (k) Thermal response of the device in the air as a function of time. (l) Optical images of the actuator operating under high and low temperatures, in water, and in sand.

IV. PATTERN EXPERIMENTS

Next, the wearable BV-actuators based VR interface installed with a number of actuators had been tested for examining the feedback fidelity and accuracy. As shown in **Fig. 4a and S20**, the wearable VR interface was directly connected to a control panel for programming and manipulating the pattern for operation. Direct current (DC) power was utilized by creating pulse width modulation (PWM) signals for inducing the electromagnetic field. To connect the device and control panel tightly and safely, a flexible printed circuit (FPC) connector was exploited and therefore, unlike wires, exceptional mechanical and electrical stability could be achieved (**Fig. 4b**). **Fig. 4b, S20, and S21** illustrate the working system of the experimental setup. A computer is connected to

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the microcontroller unit (MCU) for performing as a power source and programming the actuators while each general-purpose input/output (GPIO) of the MCU transmits a digital signal to the corresponding N-channel metal–oxide–semiconductor field-effect transistor (MOSFET) for switching. Therefore, when the MOSFETs are turned on by the MCU, an external DC power can flow to the actuators for generating vibrations. As a result, by regulating GP I/O ports through programming the MCU, different actuating patterns could be accomplished. For the accuracy tests, two wearable BV-actuator VR interfaces with different numbers and sizes of actuators and configurations (2×2 with 6.5 mm-diameter actuator and 3×3 with 3.5 mm-diameter actuator arrays) targeting a fingertip where is with the highest sensational sensitivity were employed as they are able to be fabricated in the most compact design. For each device, a total of 16 distinct patterns were prepared (Fig. 4c and 4e) and a group of 10 testers wore the devices for 10 sets of blind tests. Then, to enhance the reliability of the experiment results, the testers chose the patterns in the list they recognized from the vibrations while the preset patterns were generated randomly. Fig. 4d and 4f illustrate the testing results indicating the accuracy of pattern recognition of two devices respectively. For the 3×3 actuator array, 17 sensed patterns including offset are displayed due to the complexity of patterns resulting from the increasing number of actuators while the 2×2 device exhibits 16 sensed patterns matching with the preset patterns. According to the grid charts, the overall patterns recognition rate of two arrays (2×2 actuators array and 3×3 actuators array) could reach up to 83.75% and 80.77%, respectively. This phenomenon is on account of the high-density design of the devices and the similarities among the preset patterns. Furthermore, the devices were not customized to the testers' finger proportions, and this led to unstable adhesion between the epidermis and the device. Through the pattern recognition tests for the VR equipment, the working mechanism of the devices and the pattern accuracy of haptic feedback were demonstrated. According to the results, the requirement of customized devices for providing enhanced quality of feedback has emerged, which has been further verified in Fig. 5. Fig. S22 shows the vibration intensity level sensed by 10 volunteers, which is induced by 11-mm actuator, and it is obvious that most of volunteers could distinguish the vibration intensity with > 0.7 kPa for strong feedback, $0.3 \sim 0.7$ kPa for middle level, and < 0.3 kPa for weak level.

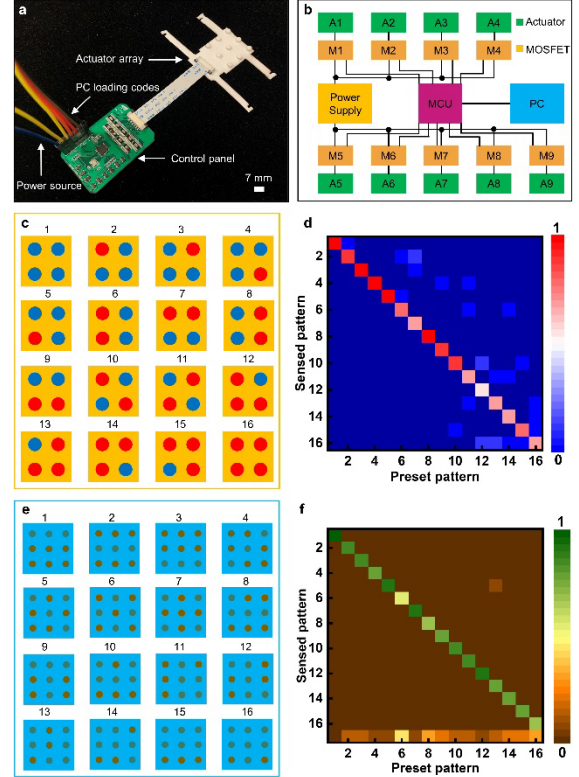


Figure 4. Programmable control and pattern actuation of the wearable VR interface. (a) Optical image of the experimental setup for controlling the device manually. (b) Block diagram of the wearable VR interface. Programmable actuating patterns for VR device with 2×2 (c) and 3×3 actuators arrays (e). Grid views of the accuracy of testing results for VR device with 2×2 (d) and 3×3 actuators arrays (f).

V. VR APPLICATIONS

The wearable VR interfaces for the forearm and hand (including palm and fingers) were next employed in a VR environment (Unity game engine) for inspecting its performance. While a user interacted with the graphical interface of the VR system, the corresponding actuators in touch or under pressure vibrated according to the applied force in the virtual world can be realized by controlling the power inputs. Therefore, a high channel of counts haptic feedback system can be achieved by manipulating each actuator individually. In the study, three different actions were tested for each device separately. To realize the high channel actuators array manipulation, we developed a printed circuit board based control panel, capable of controlling 64 actuators working separately (Fig. S23 and S24). The power consumption of the whole system could reach up to ~ 7 W as powering 64 actuators working at a full load. For instance, the forearm test associated with touching, punching, and sliding in the virtual world, where the user can feel the corresponding feeling on his arm (Fig. 5a-i). Subsequently, the vibrational force generated by each actuator represented by different colors of red, yellow, green, sky-blue, and blue can be revealed (0.72 kPa, 0.59 kPa, 0.45 kPa, 0.30 kPa, and 0 kPa applying 63.75 mA, 40.75 mA, 28.75 mA, 16.75 mA, and 0 mA of AC power, respectively). More

complicated interactions between the user and the virtual world, such as grabbing objects in a virtual world can be also realized, where it can be clearly observed that the actuators for the fingertips generated the most powerful vibrations while at the bottom of the palm was with the lowest magnitude (**Fig. 5j-o**). Similar to the aforementioned forearm device, the pressure generated by each BV-actuator was displayed in colors through the graphical distributions respectively. For instance, grabbing a ball is mainly associated with the touch between fingers and the ball, so only the actuators located on the fingertips provide an obvious mechanical force (**Fig. 5k**). Holding a cup typically uses two joints of each finger and thus detected pressure can be operated on the corresponding actuators accordingly (**Fig. 5m**). Furthermore, as the cube is placed on the palm excluding fingers, only the actuators on the palm operate according to the weight of the cube (**Fig. 5o**). Consequently, these results highly match compared to the real human motion behaviors including grabbing, being punched, poking, holding, and lifting in the aspect of force distributions of the physical motions. In addition, the average delay time when the vibration command is sent from the objects in VR games to the microcontroller of control panels for triggering the corresponding actuators is 1.3 ms, ensuring that users could smoothly interface with virtual objects in VR games. The operational performance of the wearable VR interfaces in VR environments was proved as the devices successfully reproduced the patterns resulting from VR interaction while the magnitude of pressure applied for each subregion was actualized simultaneously.

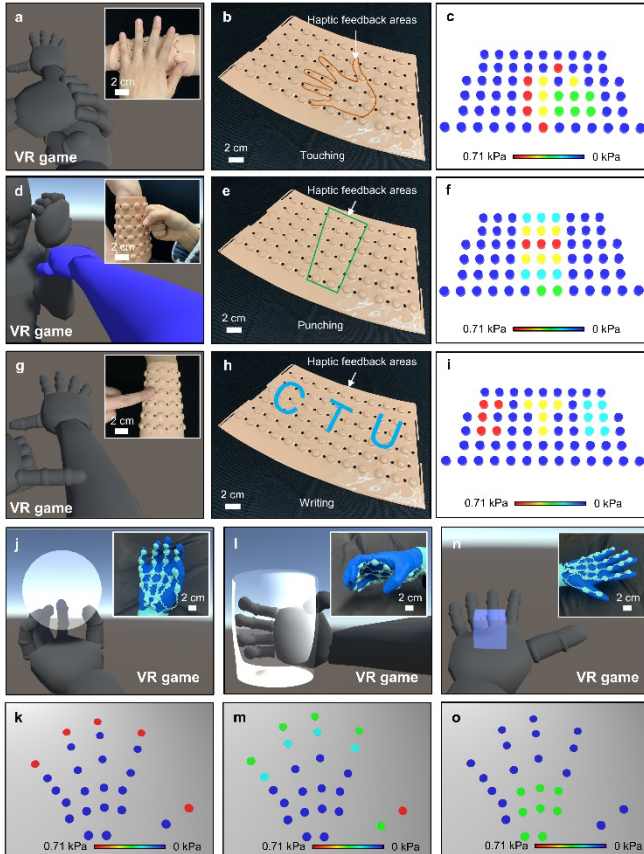


Figure 5. Response of the wearable VR interface in a VR environment. Grabbing arm in the VR interface (a) and the optical image (b) and graphical distributions (c) of haptic feedback at the device. Punching arm in the VR interface (d) and the optical image (e) and graphical distributions (f) of haptic feedback at the device. Poking the actuators in the VR interface (g) and the optical image (h) and graphical distributions (i) for producing haptic feedback of the “CTU” pattern. Grabbing a ball in the VR interface (j) and the graphical distributions (k) of haptic feedback at the device. Holding a cup in the VR interface (l) and the graphical distributions (m) of haptic feedback at the device. Placing a cube on a hand in the VR interface (n) and the graphical distributions (o) of haptic feedback at the device.

VI. CONCLUSION

In summary, a wearable low-energy electromagnetic vibrotactile actuator (BV-actuator) exploiting bilayer copper coils as the source of magnetic field for generating vibrational force is reported. Wearable VR interfaces based on the actuator arrays demonstrate enhanced haptic feedback experience in VR. The materials selection and device design allow the actuators to be configurable and exhibit robust features from mechanical tolerance to harsh environment operation, and to long term use. Through the pattern recognition tests, the accuracy and fidelity of the BV-actuator array are proven to be good enough to distinguish various patterns. In addition, its operational performance of reproducing preset patterns and actualizing the magnitude of pressure applied for each subregional actuator was verified. As a result, the BV-actuators based VR devices showed significant performance and stability while they are operated under a low-power electrical supply. Therefore, it is expected that users are able to experience vivid and dynamic haptic feedback in VR by applying equipment installed with the BV-actuators.

VII. MATERIALS AND METHODS

Fabrication of the BV-actuators. The actuators consist of six layers, including an adhesive layer, two Cu coils, two PET layers, and an epoxy ring with a magnet insides (**Fig. 1c, S25, and S26**). Here, the Cu coil was fabricated by ultrathin Cu wires with a diameter of 0.05 mm to form a concentric disc coil with a constant thickness of 0.25 mm (Yisu Electronics, Inc., Dongguan China). The inner diameter is 2 mm, and the outer diameters are 11 mm@ 11 mm actuators, 6.5 mm@ 6.5 mm actuators, and 3.5 mm@ 3.5 mm actuators, where the resistances of these Cu coils are 48 Ω , 15.7 Ω , and 3.4 Ω , respectively. The supporting epoxy rings were 3D printed at a constant thickness of 2.3 mm. Here, the outer and inner diameters are 11 mm and 9 mm@ 11 mm actuators, 6.5 mm and 5.5 mm@ 6.5 mm actuators, and 3.5 mm and 2.5 mm@ 3.5 mm actuators, where the magnet diameters of the three actuators are 8 mm, 5 mm, 2 mm, respectively. By adopting laser-cutting equipment, the 200 μ m-thick PET films could be cut into circles with three diameters, including 3.5 mm, 6.5 mm, and 11 mm. At first, a Cu coil is attached onto an adhesive layer (12/18 mm Finger Lift Double Sided Tape, Maxtape Industrial Co., Ltd),

then a PET film is bonded onto the top of the coil by quick-drying glue. After glue cured in room temperature for 5 mins, the epoxy ring is bonded onto the PET film by the quick-drying glue. Fix a magnet inside the ring, then seal the ring with a PET layer. After bonding the top Cu coil on the PET, the BV-actuator is acquired. Fabrication of the actuators array based electronics starts from the circuits fabricated by flexible printed circuit board processing techniques of copper (thickness 10 μm) plated with gold (thickness 50 nm) on PI substrate. The insulating layer was covered onto the exposing circuit for preventing short circuit. Actuators were electrically connected to the target reserved Cu patches by middle-temperature solder glue. For each actuators, four wires from the two Cu coils are combined into two sets, and each sets has two wires from the two coils. To make the coupled electromagnetic field intensity higher than that induced by a single Cu coil, the four wires are combined based on a principle where the current directions in the two coils are opposite, as shown in **Fig. 2ai**.

BV-actuator based VR interface control panel operation.

For testing the pattern recognition accuracy of the wearable BV-actuator VR equipment, a MCU (ATMEGA328P-MU, Microchip Technology Inc.) is programmed through a universal asynchronous receiver-transmitter (UART) communication exploiting wires embedded on the control panel. For 2×2 and 3×3 actuator arrays, 4 and 9 GPIOs are utilized respectively for controlling the actuators. Each digital pin from the MCU is connected to the corresponding MOSFET (SI2312BDS-T1-GE3, Vishay Intertech) for manipulating the electrical input to the actuators from an external DC power generator while the MCU is operated by a power provided from a USB port connected to a computer for programming. Therefore, when 5 V from the pin is applied to the gate of the MOSFET, the MOSFET is switched on and the current from the external source is able to flow through the corresponding actuator for operation. Furthermore, to generate vibrations, AC power is required to be applied for forcing the magnet disc up and down. Here, the MOSFET is switched on and off repeatedly for converting a DC input into a PWM signal as the AC power into the actuator (**Fig. S27**). Therefore, each digital pin is turned on and off for 8 ms respectively to achieve the electrical frequency of 62.5 Hz (highly close to the vibrational frequency for generating the maximum haptic feedback, 60 Hz). For each device, 16 preset patterns are prepared and to promote the patterns, the MCU is programmed for generating PWM signals to the actuators required to be vibrated while the GPIOs for the actuators excluded in the patterns are turned off.

High-channel control panel operation. For operating a high-channel actuator system that employs 64 channels in total for generating corresponding actuators independently as a test, 64 digital GPIOs of a MCU (STM32F407IGT6, STMicroelectronics Inc.) are exploited to create 60 Hz of PWM signals with different duty ratios for separated channels concurrently. The whole system is managed by 3.3 V while the reference voltage for actuators is supplied by an external DC power source. For generating actuators, two different methods are applied in one circuit system which are using the combination of a decoder (SN74LVC1G19DBVR, Texas

Instruments Inc.) and H-Bridge (BD6211F-E2, ROHM Semiconductor) and MOSFET (AO3400A, Alpha & Omega Semiconductor Inc.). Firstly, the decoder that outputs two opposite levels of signals which one is high (3.3 V) while the other is low (0 V) and vice versa according to a single input signal (high or low) is used to reduce the number of pins adopted for each channel as H-Bridge itself requires two opposite levels of input signals for managing the forward and reverse functions. As a result, by exploiting the PWM signal from a GPIO of the MCU, the decoder generates opposite signals based on the PWM duty ratio and therefore, the H-Bridge is able to shift the electrical current direction of the reference voltage from the external DC power source for mimicking AC characteristics on an actuator side. Consequently, the target actuator is generated with positive and negative voltage resulting from the H-Bridge in 60 Hz with a peak-to-peak voltage that is doubled the reference voltage. Secondly, by applying the 60 Hz of PWM signal from a GPIO of the MCU to the gate of MOSFET, the MOSFET is turned on and off accordingly. Therefore, electrical current from the external DC power source can flow to an actuator at 60 Hz. Thus, the corresponding actuator vibrates 60 times per second. Similarly, the magnitudes of individual actuators are regulated by either the PWM duty ratio or the value of the reference voltage in both methods. Practically, the magnitudes of graphical contact for each actuator in the VR system would be converted into 1-byte digital data and in total 64 bytes of data would be transmitted to the MCU through UART serial communication from a computer. Thereafter, in accordance with the data transmitted, the PWM duty ratios for separated actuators would be adjusted for independent operation. Consequently, 64 actuators can independently regenerate corresponding magnitudes of contact in the form of vibration.

VR game operation. The VR game is built by the commercial software, Unity. On the humanoid 3D model, we place invisible collision detection points on the model's surface. The amount and location of the collision detection points are the same as the actuators on our devices, each collision detection points have its own collider which is able to detect collision with another collider. When the humanoid model is in contact with another 3D object (e.g. a cup), the program in Unity will detect whether the collision detection points have contact with the cup's collider, then calculate the penetration depth between the collider of collision detection points and cup's collider. If two colliders are overlapped and have a positive penetration depth, we consider the collision detection point is contact with the target object, command to turn on the corresponding actuator on our device will be delivered via USB port using serial communication. In addition, the VR vision can be provided by the Oculus quest headset and the Oculus development library in Unity.

Fabrication of the flexible piezoelectric pressure sensors.

The piezoceramic thin film used in this work is an emerging lead-free material $\text{Ba}_{0.85}\text{Ca}_{0.15}\text{Zr}_{0.1}\text{Ti}_{0.9}\text{O}_3$ (BCZT) with a d_{33} of 209 pC/N. The 400nm-thick BCZT thin film is prepared based on mica sheets via sol-gel technique and the detailed procedures have been described in our previous work[31]. As

shown in **Fig. S28**, the mica substrate is mechanically exfoliated by tapes to reduce the thickness and the BCZT thin film is then transferred to the Pt-coated PDMS thin film, which is served as the bottom electrode. The top electrode is magnetron sputtering (Q150TS, Quorumtech) derived Pt thin film and encapsulated by a thin layer of PDMS (Young's modulus, 147 kPa). The piezoceramic thin film is polarized under 100 kV/cm at room temperature. The cross-sectional view of the BCZT film is examined by scanning electron microscope (SEM, Quanta FEI 450), as shown in **Fig. S29a**. The thickness of the BCZT thin film is about 400 nm and the mica sheets are reduced to approximately 1 μm . The crystallinity of the BCZT thin film is investigated by X-ray diffraction (XRD, Rigaku Smartlab) and Raman spectroscopy (Perkinelmer Raman station 400F) and the results are illustrated in **Fig. S29b, c**. The XRD spectrum shows the typical perovskite phase and the Raman spectrum exhibit the tetragonal phase of the film. Here, we used a commercial oscillator with the pressures measured by a commercial pressure sensor (SBT674, Guangzhou Simbatouch Electronic Technology Co., LTD) in a low working frequency, which is used to measure the electrical response of the self-developed sensor as a function of the applied pressure (**Fig. S11b**).

Mechanical analysis. The FEA commercial software ABAQUS (Analysis User's Manual 2020) was used to obtain the impact resistance capacity of actuator. The Cu, PET, Epoxy and Magnet were modelled by 0.86 million hexahedron elements (C3D8R). The minimal element size was 0.002mm, which ensured the convergence and the accuracy of the simulation results. The uniformly distributed force with pressure of 5.53MPa was applied on the top surface of the actuator and the magnet is set as rigid body. The elastic modulus (E) and Poisson's ratio (ν) used in the analysis were $E_{\text{Cu}}=131$ GPa, $\nu_{\text{Cu}}=0.33$, $E_{\text{PET}}=3.5$ GPa, $\nu_{\text{PET}}=0.35$, $E_{\text{Epoxy}}=3.35$ GPa, $\nu_{\text{Epoxy}}=0.35$.

Thermal analysis. The actuator was modelled in ABAQUS (Analysis User's Manual 2020) and eight-node linear heat-transfer element DC3D8 were adopted to discretize the geometry. The grid size was constrained in an appropriate range of 0.002 and 0.1mm with the total number of grids being about 0.97 million, which ensured both acceptable convergence and computational efficiency. The thermal conductivity of Cu, PET, Epoxy, Magnet, dry sand and water are $394 \text{ Wm}^{-1}\text{K}^{-1}$, $0.3 \text{ Wm}^{-1}\text{K}^{-1}$, $0.2 \text{ Wm}^{-1}\text{K}^{-1}$, $9 \text{ Wm}^{-1}\text{K}^{-1}$, $0.15 \text{ Wm}^{-1}\text{K}^{-1}$ and $0.598 \text{ Wm}^{-1}\text{K}^{-1}$. The convective heat transfer coefficients are selected as $h_{\text{air}} = 25 \text{ Wm}^{-2}\text{K}^{-1}$ in air and $h_{\text{water}} = 200 \text{ Wm}^{-2}\text{K}^{-1}$ in water.

APPENDIX

All data are contained within the manuscript. Raw data are available from the corresponding authors upon reasonable request.

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AUTHOR CONTRIBUTIONS

Y.L., C.Y., Z.Z., S.L., and X.H. contributed equally to this work. Y.L., X.Y., Y.L., and Z.Y. conceived the ideas, and designed the experiments. Y.L., W.P., T.W., and X.Y. wrote the manuscript. Y.L., W.P., T.W., J.S., X.H., K.Y., L.Z., Y.H., B.Z., Y.D., P.F., and J.Z. performed experiments and analyzed the experimental data. W.P., J.Z., and C.Y. performed electrical circuit designs. Z.Z. and Y.L. performed the theoretical mechanical and thermo analysis. S.L. and Z.Y. performed the piezoelectric pressure sensor fabrication.

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